Atiyah-Hirzebruch Integral Hodge Counterexamples and Torsion Obstructions: Formal Separation from the Rational Clay Hodge Conjecture

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Abstract

Standalone formal documentation and mathematical ledger of the 1962 Atiyah-Hirzebruch torsion counterexamples to the Integral Hodge Conjecture on smooth projective varieties over C, constructed via Steenrod operations in topological K-theory on Godeaux-Serre quotients of products of elliptic curves by elementary abelian p-groups (p >= 3) in complex dimension >= 7. Explicitly documents why torsion classes vanish under tensoring with Q (H^{2p}(X, Z)_tors (x) Q = 0) and therefore do not constitute counterexamples to the Clay Mathematics Institute Rational Hodge Conjecture.

1 Scope, Scientific Honesty, and Non-Claim Boundary

This repository records known 1962 integral torsion obstructions (Atiyah-Hirzebruch) and their vanishing after rationalization. It does not prove or disprove the rational Clay Hodge Conjecture.

- **Repository:** `https://github.com/BenFrohman/AtiyahHirzebruch`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23652971`
- **Earliest Git Commit SHA:** `4324e2550e8982bf1555a837ef0ead5957e00e26` (2026-09-28 06:55:02 PM CDT (UTC-5))
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2 Technical Overview & Repository Specification

Status: literature counterexample to the *integral* Hodge conjecture.
Not a counterexample to Clay rational Hodge.
Author: Benjamin Stanley Frohman (@BenFrohman)
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This shelf records one of the two classical counterexamples to
“‘
CH^k(X) → H^{2k}(X,Z) H^{k,k}(X)
“‘
being surjective. The other shelf is Kollár (non-torsion). Neither shelf is Term B of rational Hodge.

There exist smooth complex projective varieties and torsion classes

‘‘‘

H^{2k}(X,Z) H^{k,k}(X), n = 0 for some n > 0,

’’’

that are not the class of an algebraic cycle.

Source: M. F. Atiyah and F. Hirzebruch, *Analytic cycles on complex manifolds*, Topology 1 (1962), 25–45.

Totaro (1997) recast the same examples in complex cobordism: the refined cycle class ‘CH^*(X) → MU^*(X) _{MU_*} Z‘ misses those torsion classes.

Clay Hodge is the *rational* statement:

‘‘‘

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/AtiyahHirzebruch` is audited via `titan-audit` (`audit/TITAN_AUDIT_CER` and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

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